

MS Fabric Finance Ad Hoc Analytics

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1  -- SQL Query (Finance) Analysis:
2
3  SELECT * FROM dim_customer;
4
5  SELECT * FROM fct_transactions;
6
7  -- 1. Total Transactions: 50,000
8
9  SELECT COUNT(*) AS total_transactions
10 FROM fct_transactions;
11
12 -- 2. Total Customers: 5000
13
14 SELECT COUNT(*) AS total_customers
15 FROM dim_customer;
16
17 -- 3. Average Customer Age: 44 years
18
19 SELECT AVG(age) AS avg_customer_age
20 FROM dim_customer;
21
22 -- 4. Total Transaction Amount:Rs. 455.53 Million
23
24 SELECT CONCAT(ROUND((SUM(amount))/100000, 2), ' M') AS total_transaction_amt
25 FROM fct_transactions;
26
27 -- 5. Total Fees Collected: Rs.7,26,241.72
28
29 SELECT ROUND(SUM(fee_amount), 2) AS total_fees
30 FROM fct_transactions;
31
32 -- 6. Total Tax Amount: Rs.130736.35
33
34 SELECT ROUND(SUM(tax_amount),2) AS total_tax
35 FROM fct_transactions;
36
37 -- 7. Average Transaction Amount: Rs.9110.68
38
39 SELECT ROUND(AVG(amount),2) AS avg_transaction_amount
40 FROM fct_transactions;
41
42 -- 8. Average Annual Income: Rs.724095.28
43
44 SELECT ROUND(AVG(annual_income),2) AS avg_annual_income
45 FROM dim_customer;
46
47 -- 9. Successful Transactions: 42869
48
49 SELECT COUNT(*) AS successful_transactions
50 FROM fct_transactions
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51 WHERE transaction_status = 'Success';
52
53 -- 10. Fraud Amount: Rs.59,69,443.52
54
55 SELECT SUM(amount) AS fraud_amount
56 FROM fct_transactions
57 WHERE is_fraud = 'Yes';
58
59 -- 11. Success Rate %
60
61 SELECT CAST(
62     COUNT(CASE WHEN transaction_status = 'Success' THEN 1 END) * 100.0
63     / COUNT(*) AS DECIMAL(10,2)
64 ) AS success_rate_perc
65 FROM fct_transactions;
66
67 -- 12. Fraud Rate %
68
69 SELECT CAST(
70     COUNT(CASE WHEN is_fraud = 'Yes' THEN 1 END) * 100.0
71     / COUNT(*) AS DECIMAL(10,2)
72 ) AS fraud_rate_percentage
73 FROM fct_transactions;
74
75 -- 13. Granular Insights:
76
77 -- 13.1 Total Monthly Transaction Amount:
78
79 SELECT
80     YEAR(transaction_date) AS year_,
81     MONTH(transaction_date) AS month_no,
82     DATENAME(MONTH, transaction_date) AS month_nm,
83     ROUND(SUM(amount),2) AS total_amount
84 FROM fct_transactions
85 GROUP BY
86     YEAR(transaction_date),
87     MONTH(transaction_date),
88     DATENAME(MONTH, transaction_date)
89 ORDER BY year_, month_no;
90
91 -- 13.2 Transaction Status Distribution:
92
93 SELECT transaction_status, COUNT(*) AS total_transactions
94 FROM fct_transactions
95 GROUP BY transaction_status;
96
97 -- 13.3 Transaction Type Analysis:
98
99 SELECT
100     transaction_type,
101     COUNT(*) AS total_transactions,
102     ROUND(SUM(amount),2) AS total_amount
103 FROM fct_transactions
104 GROUP BY transaction_type

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105 ORDER BY total_amount DESC;
106
107 -- 14. Customers by State:
108
109 SELECT state, COUNT(*) AS total_customers FROM dim_customer
110 GROUP BY state
111 ORDER BY total_customers DESC;
112
113 -- 15. Customers by Occupations:
114
115 SELECT occupation, COUNT(*) AS total_customers FROM dim_customer
116 GROUP BY occupation
117 ORDER BY total_customers DESC;
118
119 -- 16. Customer Segment Distribution:
120
121 SELECT customer_segment, COUNT(*) AS total_customers
122 FROM dim_customer
123 GROUP BY customer_segment;
124
125 -- 17. Age Group Analysis:
126
127 SELECT CASE
128     WHEN age < 25 THEN 'Below 25'
129     WHEN age BETWEEN 25 AND 40 THEN '25-40'
130     WHEN age BETWEEN 41 AND 60 THEN '41-60'
131     ELSE '60+'
132 END AS age_group, COUNT(*) AS total_customers
133 FROM dim_customer
134 GROUP BY
135     CASE
136     WHEN age < 25 THEN 'Below 25'
137     WHEN age BETWEEN 25 AND 40 THEN '25-40'
138     WHEN age BETWEEN 41 AND 60 THEN '41-60'
139     ELSE '60+'
140 END
141 ORDER BY total_customers DESC;
142
143 -- 18. Top 5 Highest Transactions:
144
145 SELECT TOP 5 transaction_id, customer_id, amount, transaction_date
146 FROM fct_transactions
147 ORDER BY amount DESC;
148
149 -- 19. Top 10 Customers by Transaction Amount:
150
151 SELECT TOP 10 c.full_name, ROUND(SUM(f.amount),2) AS total_transaction_amount
152 FROM fct_transactions f
153 INNER JOIN dim_customer c ON f.customer_id = c.customer_id
154 GROUP BY c.full_name
155 ORDER BY total_transaction_amount DESC;
156
157 -- 20. Monthly Revenue by Customer Segment:
158

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159 SELECT
160     YEAR(f.transaction_date) AS year_,
161     MONTH(f.transaction_date) AS month_,
162     c.customer_segment,
163     ROUND(SUM(f.fee_amount + f.tax_amount),2) AS revenue
164 FROM fct_transactions f
165 INNER JOIN dim_customer c ON f.customer_id = c.customer_id
166 GROUP BY
167     YEAR(f.transaction_date),
168     MONTH(f.transaction_date),
169     c.customer_segment
170 ORDER BY year_, month_;
171
172
173 -- 21. Total Transaction Amount by Gender:
174
175 SELECT c.gender, ROUND(SUM(f.amount),2) AS total_transaction_amt
176 FROM fct_transactions f
177 INNER JOIN dim_customer c ON f.customer_id = c.customer_id
178 GROUP BY c.gender
179 ORDER BY total_transaction_amt DESC;
180
181 -- 22. Revenue by Customer Segment:
182
183 SELECT c.customer_segment, ROUND(SUM(f.fee_amount + f.tax_amount),2) AS revenue
184 FROM fct_transactions f
185 INNER JOIN dim_customer c ON f.customer_id = c.customer_id
186 GROUP BY c.customer_segment
187 ORDER BY revenue DESC;
188
189 -- 23. Top States by Transaction Amount:
190
191 SELECT c.state, ROUND(SUM(f.amount),2) AS total_transaction_amount
192 FROM fct_transactions f
193 INNER JOIN dim_customer c ON f.customer_id = c.customer_id
194 GROUP BY c.state
195 ORDER BY total_transaction_amount DESC;
196
197 -- 24. Top 6 Cities by Revenue:
198
199 SELECT TOP 6 c.city, ROUND(SUM(f.fee_amount + f.tax_amount),2) AS revenue
200 FROM fct_transactions f
201 INNER JOIN dim_customer c ON f.customer_id = c.customer_id
202 GROUP BY c.city
203 ORDER BY revenue DESC;
204
205 -- 25. Age Group Wise Revenue:
206
207 SELECT CASE
208     WHEN c.age < 25 THEN 'Below 25'
209     WHEN c.age BETWEEN 25 AND 40 THEN '25-40'
210     WHEN c.age BETWEEN 41 AND 60 THEN '41-60'
211     ELSE '60+'
212     END AS age_group,

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213
214 ROUND(SUM(f.fee_amount + f.tax_amount),2) AS revenue
215 FROM fct_transactions f
216 INNER JOIN dim_customer c ON f.customer_id = c.customer_id
217
218 GROUP BY
219     CASE
220         WHEN c.age < 25 THEN 'Below 25'
221         WHEN c.age BETWEEN 25 AND 40 THEN '25-40'
222         WHEN c.age BETWEEN 41 AND 60 THEN '41-60'
223         ELSE '60+'
224     END
225 ORDER BY revenue DESC;
226
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